K.R. MANGALAM UNIVERSITY

THE COMPLETE WORLD OF EDUCATION

School of Engineering & Technology

OPERATING SYSTEM

LAB ASSINGMENT 4

Program: BCA (AI & DS)

Course Code – ENCA252 (2026-2027)

Submitted By	Submitted To
RAKESH G	JYOTI YADAV
2401201064	
BCA (AI & DS)	

Assignment Tasks – Output

Task 1: Input and Disk Request Representation

Task 2: FCFS Disk Scheduling

Task 3: SSTF (Shortest Seek Time First)

Task 4: SCAN (Elevator Algorithm)

Task 5: C-SCAN Algorithm

```
rakesh@rakesh-VirtualBox:~/Documents$ python3 disk_scheduling.py

============================================================
        DISK SCHEDULING ALGORITHM SIMULATOR
============================================================

Enter initial head position: 53
Enter disk size (total cylinders): 200
Enter disk request queue (space-separated): 98 183 37 122 14 124 65 67

Initial Head  : 53
Disk Size     : 200
Request Queue : [98, 183, 37, 122, 14, 124, 65, 67]
Total Requests: 8

────────────────────────────────────────────────────────────
  FCFS  (First Come First Serve)
────────────────────────────────────────────────────────────
  Head Movement : 53 → 98 → 183 → 37 → 122 → 14 → 124 → 65 → 67
  Total Seek Time: 640 cylinders

────────────────────────────────────────────────────────────
  SSTF  (Shortest Seek Time First)
────────────────────────────────────────────────────────────
  Head Movement : 53 → 65 → 67 → 37 → 14 → 98 → 122 → 124 → 183
  Total Seek Time: 236 cylinders

────────────────────────────────────────────────────────────
  SCAN  (Elevator Algorithm)
────────────────────────────────────────────────────────────
  Head Movement : 53 → 65 → 67 → 98 → 122 → 124 → 183 → 199 → 37 → 14
  Total Seek Time: 331 cylinders

────────────────────────────────────────────────────────────
  C-SCAN (Circular SCAN)
────────────────────────────────────────────────────────────
  Head Movement : 53 → 65 → 67 → 98 → 122 → 124 → 183 → 199 → 0 → 14 → 37
  Total Seek Time: 382 cylinders

============================================================
           PERFORMANCE COMPARISON
============================================================
Algorithm            |  Total Seek Time | Rank
────────────────────────────────────────────────────────
SSTF                 |              236 cylinders | #1
SCAN                 |              331 cylinders | #2
C-SCAN               |              382 cylinders | #3
FCFS                 |              640 cylinders | #4

  ✔ Best  : SSTF  (236 cylinders)
```

Linux [Running] - Oracle VirtualBox

File Machine View Input Devices Help

```
  Head Movement : 53 → 65 → 67 → 98 → 122 → 124 → 183 → 199 → 0 → 14 → 37
  Total Seek Time: 382 cylinders

============================================================
           PERFORMANCE COMPARISON
============================================================
Algorithm            |  Total Seek Time | Rank
____________________________________________________________
SSTF                 |              236 cylinders | #1
SCAN                 |              331 cylinders | #2
C-SCAN               |              382 cylinders | #3
FCFS                 |              640 cylinders | #4

  ✔ Best  : SSTF  (236 cylinders)
  ✘ Worst : FCFS (640 cylinders)

============================================================
           RESULT ANALYSIS & CONCLUSION
============================================================

  [FCFS]  —  640 cylinders
    Simple and fair — requests served in arrival order.
    Con: High seek time. No optimization. Poor performance.

  [SSTF]  —  236 cylinders
    Always picks the nearest request — reduces seek time.
    Con: Can cause starvation for far-away requests.

  [SCAN]  —  331 cylinders
    Moves in one direction, reverses at end — like an elevator.
    Pro: Balanced. No starvation. Good overall performance.

  [C-SCAN]  —  382 cylinders
    One-directional only, jumps back to 0 after reaching end.
    Pro: Uniform wait time. Fairer than SCAN for all requests.

  CONCLUSION:
  • SSTF usually gives minimum seek time but risks starvation.
  • SCAN is the most balanced for real-world use.
  • C-SCAN gives uniform response — best for heavy load systems.
  • FCFS is simplest but least efficient.

  In this simulation, SSTF performed best with 236 cylinders of movement.

============================================================
  Simulation Complete!
============================================================

rakesh@rakesh-VirtualBox:~/Documents$
```